# STRONG FORMAL EFFECTIVENESS FOR THE STACK OF RELATIVELY PERFECT COMPLEXES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5847. Standing situation: B = S locally Noetherian.
- Remark artin-remark-strong-effectiveness (canonical tag 0CXT). cited at statement line 5849. The property being established: strong formal effectiveness.
- Lemma lemma-complexes (canonical tag 0DLD). cited at proof lines 5866 and 5884. Condition (2) for the objects and for the limit.
- Theorem spaces-flat-theorem-existence-derived (canonical tag 0DIQ). cited at proof line 5870. Existence of E with the prescribed derived pullbacks.
- Lemma more-algebra-lemma-limit-henselian (canonical tag 0CT7). cited at proof line 5877. (R, Ker(R -> R_1)) is a henselian pair.
- Lemma spaces-more-morphisms-lemma-henselian-relatively-perfect (canonical tag 0DL5). cited at proof lines 5879-5880. E is R-perfect.
- Lemma lemma-complexes-open-neg-exts-vanishing (canonical tag 0DLC). cited at proof line 5885. The negative-Ext vanishing locus is open.

*Source reference.* Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5847: the lemma is phrased in this situation; load-bearing.

*Source reference.* Original remark 0CXT: sources/stacks-chow/tags.txt, source lines 16218–16218. Cited at statement line 5849: the property being established; load-bearing.

*Source reference.* Original lemma 0DLD: sources/stacks-chow/tags.txt, source lines 17043–17043. Cited at proof lines 5866 and 5884: condition (2) for the objects and for the limit; load-bearing twice.

*Source reference.* Original theorem 0DIQ: sources/stacks-chow/tags.txt, source lines 16950–16950. Cited at proof line 5870: the object E of D(O_{X_R}) with E_n its derived pullbacks; load-bearing.

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*Date*: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0CT7: sources/stacks-chow/tags.txt, source lines 16057–16057. Cited at proof line 5877: the henselian pair, giving Ker in the Jacobson radical; load-bearing.

*Source reference.* Original lemma 0DL5: sources/stacks-chow/tags.txt, source lines 17035–17035. Cited at proof lines 5879-5880: E is R-perfect; load-bearing.

*Source reference.* Original lemma 0DLC: sources/stacks-chow/tags.txt, source lines 17042–17042. Cited at proof line 5885: the negative-Ext vanishing locus is open; load-bearing.

**Lemma 0.2.** *In Situation 0DLE assume $B = S$ is locally Noetherian. Then strong formal effectiveness in the sense of Artin's Axioms, Remark 0CXT holds for $p : \mathcal{C}omplexes_{X/S} \to (Sch/S)_{fppf}$.*

*Proof.* Let $(R_n)$ be an inverse system of $S$-algebras with surjective transition maps whose kernels are locally nilpotent. Set $R = \lim R_n$. Let $(\xi_n)$ be a system of objects of $\mathcal{C}omplexes_{X/B}$ lying over $(\mathrm{Spec}(R_n))$. We have to show $(\xi_n)$ is effective, i.e., there exists an object $\xi$ of $\mathcal{C}omplexes_{X/B}$ lying over $\mathrm{Spec}(R)$.

Write $X_R = \mathrm{Spec}(R) \times_S X$ and $X_n = \mathrm{Spec}(R_n) \times_S X$. Of course $X_n$ is the base change of $X_R$ by $R \to R_n$. Since $S = B$, we see that $\xi_n$ corresponds simply to an $R_n$-perfect object $E_n \in D(\mathcal{O}_{X_n})$ satisfying condition (2) of Lemma 0DLD. In particular $E_n$ is pseudo-coherent. The isomorphisms $\xi_{n+1}|_{\mathrm{Spec}(R_n)} \cong \xi_n$ correspond to isomorphisms $L(X_n \to X_{n+1})^* E_{n+1} \to E_n$. Therefore by Flatness on Spaces, Theorem 0DIQ we find a pseudo-coherent object $E$ of $D(\mathcal{O}_{X_R})$ with $E_n$ equal to the derived pullback of $E$ for all $n$ compatible with the transition isomorphisms.

Observe that $(R, \mathrm{Ker}(R \to R_1))$ is a henselian pair, see More on Algebra, Lemma 0CT7. In particular, $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical of $R$. Then we may apply More on Morphisms of Spaces, Lemma 0DL5 to see that $E$ is $R$-perfect.

Finally, we have to check condition (2) of Lemma 0DLD. By Lemma 0DLC the set of points $t$ of $\mathrm{Spec}(R)$ where the negative self-exts of $E_t$ vanish is an open. Since this condition is true in $V(\mathrm{Ker}(R \to R_1))$ and since $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical of $R$ we conclude it holds for all points. □